

Mini Project Presentation

AI BASED DIGITAL CONTENT ORGANIZER

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Sem: V

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SYMBIOSIS INTERNATIONAL (DEEMED UNIVERSITY)

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<Relevant heading of the slide>

1. Presentation Flow



Problem Statement

Section 01



The rapid growth of digital content makes it difficult to manually identify, categorize, and organize useful videos.



Large volume of online content



Manual categorization is time-consuming



Difficult to maintain organized collections



Search retrieves content but does not organize it into predefined categories



Need for an automated content organization system

Research Initiatives / Objectives

Section 02



PRIMARY GOAL

**Automate digital
content
organization**



Retrieve real-time public video content



Classify content using
TF-IDF + Logistic Regression



Categorize videos into 8
predefined categories



Display prediction confidence



Provide an easy-to-use Gradio
interface



Maintain a personal content
library

Existing Processes / Solutions

Section 03



Manual Bookmarking & Organization

Users save and sort links themselves

LIMITATION

Requires continuous manual effort



Basic Platform Search

Returns results that match a query

LIMITATION

Focuses mainly on relevance, not category organization



Recommendation Systems

Suggests content tailored to a user

LIMITATION

Focus on personalization rather than explicit categories



Traditional Text Classification

Assigns text to predefined classes

LIMITATION

May not integrate real-time retrieval or personal library management

Gap: automatic categorization + real-time retrieval + personal library in one workflow

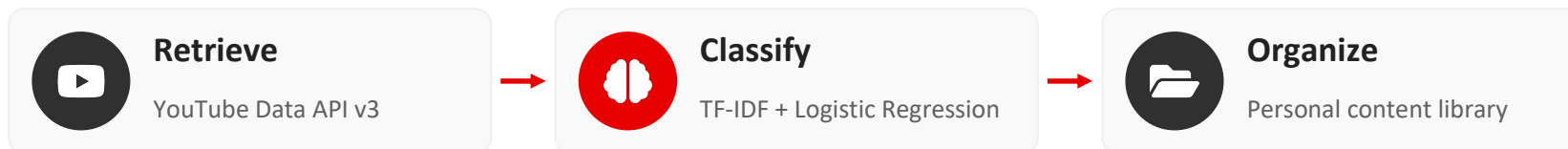
Compare & Contrast Alternative Solutions

Section 04

Approach	Main Purpose	Automatic Classification	Real-Time Retrieval	Personal Library
Manual Organization	Organize saved content by hand	No	No	Yes (manual)
Search Systems	Retrieve content relevant to a query	No	Yes	No
Recommendation Systems	Suggest personalized content	Partial	Varies	No
Traditional Text Classification	Assign text to predefined classes	Yes	Not typical	Not typical
Proposed System	Retrieve, classify & organize videos	Yes	Yes	Yes

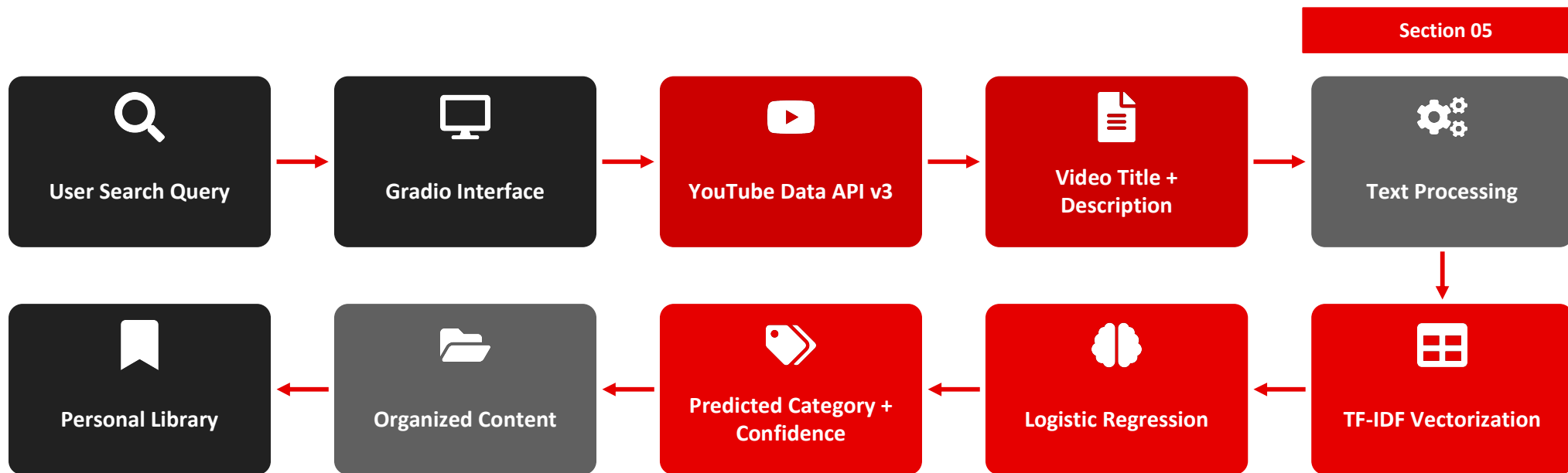
Comparison reflects the general purpose of each approach; it is not a performance rating.

PROPOSED WORKFLOW



Problem Modeling and Algorithm

Section 05



TF-IDF

Converts text into numerical features.

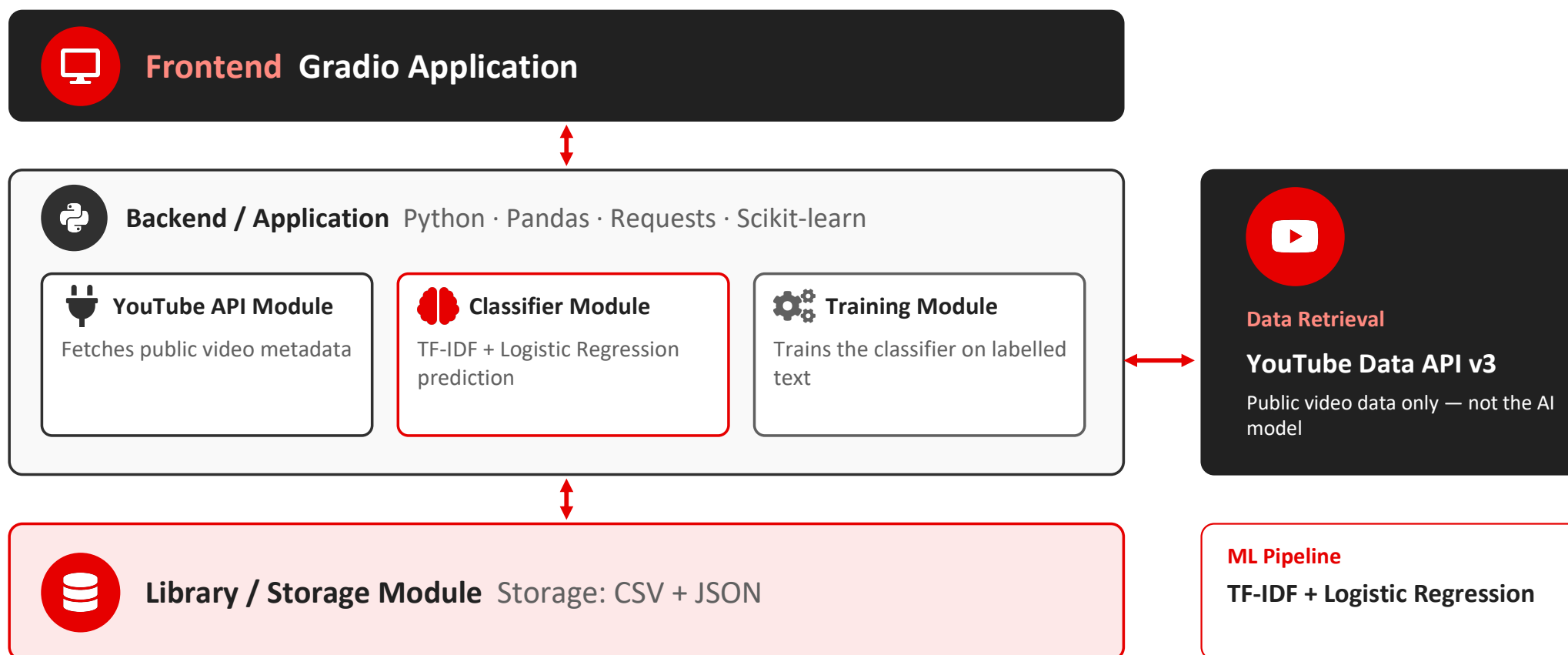


Logistic Regression

Performs supervised classification into the 8 categories.

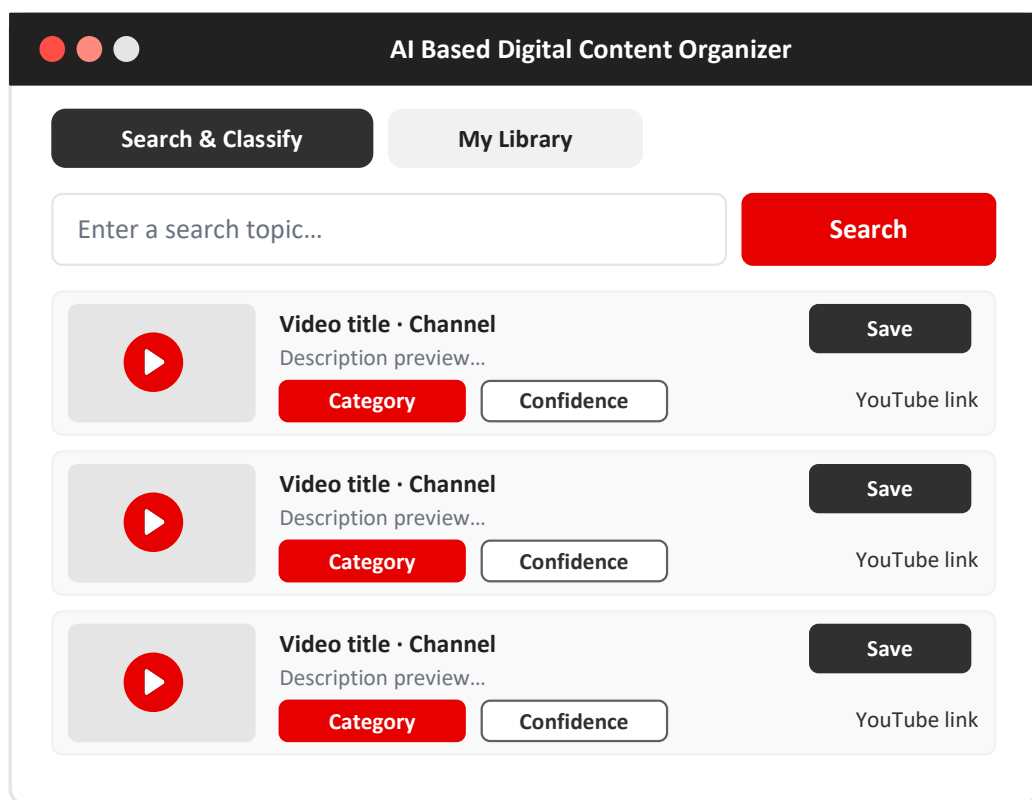
Development: Architecture & Modules

Section 05



Implementation of Project Features

Section 06



Illustrative UI layout (not an actual screenshot)



Search YouTube content



Retrieve real video metadata



AI-based classification



Category-wise organization



Confidence score



Thumbnails and links



Personal content library



Search / filter library



CSV / JSON storage

Results and Outcomes

Section 07

MODEL EVALUATION

97.50%

Test accuracy

TF-IDF + Logistic Regression

Evaluated on a 40-sample test split

Example / demo run — not universal performance

Search query: “Python Programming”

10

Videos retrieved

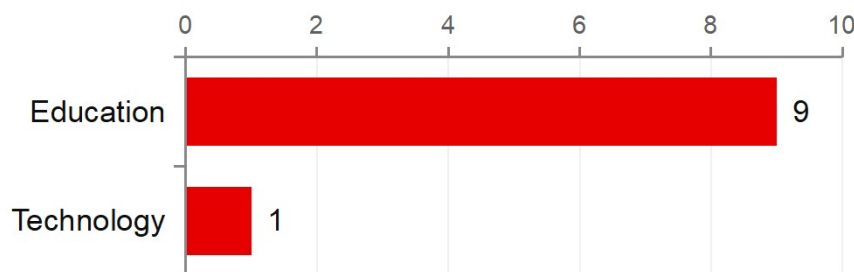
2

Categories detected: Education,
Technology

≈29.9%

Average AI confidence (this run)

Category distribution (videos)



Reading the demo

Distribution shown is from a single query. It will vary with the search topic and the videos retrieved.

Analysis of Developed Solution

Section 08



Strengths

- Automated categorization
- Real-time YouTube content retrieval
- Simple user interface
- Lightweight ML approach
- Confidence score for each prediction
- Personal content library
- Modular architecture



Limitations

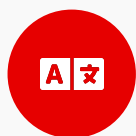
- Classification depends on title and description quality
- Limited to predefined categories
- Text-only: does not analyze actual video/audio content
- API availability / quota can affect retrieval
- Low-confidence predictions can occur for ambiguous content

Future Scope

Section 09



**Transformer / BERT
classification**



**Multilingual
classification**



**Additional content
categories**



**Support for more
content platforms**



**Caption / tag-based
analysis**



**Personalized
recommendations**



Cloud deployment



**Advanced search
and filtering**



Mobile application



**Continuous model
retraining**

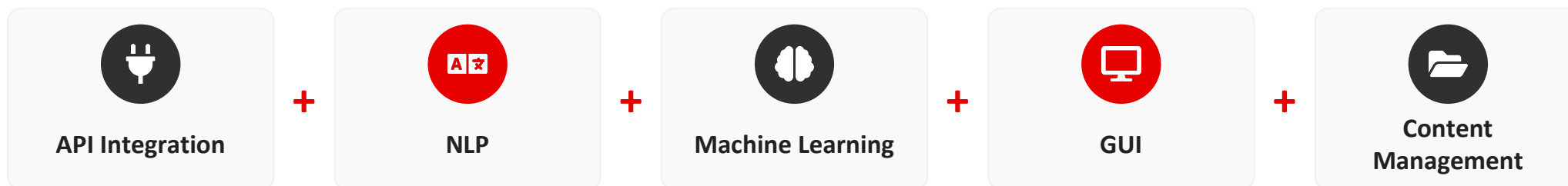
Conclusion

Section 10

The AI Based Digital Content Organizer integrates real-time YouTube content retrieval with machine learning-based text classification.

TF-IDF and Logistic Regression classify retrieved videos into predefined categories, while Gradio provides an interactive interface for viewing and organizing results.

The project demonstrates the practical integration of



References

Section 11

1

YouTube Data API Documentation

Google for Developers — developers.google.com/youtube/v3

2

YouTube Data API: Search: list

developers.google.com/youtube/v3/docs/search/list

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Scikit-learn Documentation

scikit-learn.org/stable

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TfidfVectorizer — Scikit-learn Documentation

scikit-learn.org/stable/modules/generated/sklearn.feature_extraction.text.TfidfVectorizer.html

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LogisticRegression — Scikit-learn Documentation

scikit-learn.org/stable/modules/generated/sklearn.linear_model.LogisticRegression.html

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Gradio Documentation

gradio.app/docs

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Sebastiani, F. — Machine Learning in Automated Text Categorization

ACM Computing Surveys, 34(1), 2002